



ACADEMIC QUALIFICATIONS			
Year	Degree	University/Institution	Location
2022 – 2026	B.Tech in CSE; Honors in AI & ML	Manipal University Jaipur	Jaipur, RJ
WORK EXPERIENCE			
Graduate Engineering Trainee		Navitasys India Pvt. Ltd.	Manesar, HR (Jan '25 – Present)
Pipeline & Infrastructure	- Engineered a two-stage anomaly detection pipeline (XGBoost regression + binary NG classifier) over a 600M+ row, 790-shard+ Parquet dataset spanning battery manufacturing test records. - Architected memory-efficient streaming ingestion using FactoryDataIter / QuantileDMatrix with temporally-correct 85/15 shard-level train-val splits and globally consistent categorical encodings, eliminating data leakage and OOM failures on a 128 GB RAM workstation.		
Computer Vision & Applications	- Designed and deployed a security computer vision system for factory anomaly detection, building a multi-stage inference pipeline of a custom-trained YOLO model coupled with downstream LLM and vision-language models for contextual classification and automated alerting. - Built an HR chatbot backend (Flask, Gemini 2.0 Flash) with automatic function calling for leave management, and migrated SPC and OEE dashboards to FastAPI with Shewhart rule violation detection and real-time C_P/C_{pk} computation.		
Engineering Manager		Carbon Crunch	Hybrid (Sept '24 – Jan '25)
Roles & Responsibilities	- Architected enterprise-wide carbon emissions system tracking 20+ data sources across Scope 1 & 2 emissions, reducing manual reporting time by 90% through automated data integration pipelines. - Led development of carbon credit trading platform serving 7+ corporate clients, implementing real-time pricing engine and automated compliance verification. - Engineered ETL pipelines processing 10,000+ financial and environmental reports through company internal ecosystems, utilizing extracted data to enforce compliance guidelines.		
Defense Comm. System Architect		Indian Defense R&D Project	Army Bases Nationwide
Roles & Responsibilities	- Designed inter-device communication framework for defense systems, architecting secure protocols for mission-critical military operations across distributed networks. - Conducted technical assessments at multiple army bases nationwide, gathering requirements from defense personnel to develop architectural guidelines for indigenous equipment R&D modernization. - Currently in process of securing funding grant from Army Design Bureau for full-scale deployment of communication framework.		
Technical Instructor		Camp K12	Remote (Jan '23 – Apr '23)
Roles & Responsibilities	Delivered Python OOP curriculum to 30+ students (grades 8-12) achieving 90% course completion rate through customized teaching methodologies and hands-on project-based learning.		
PROJECTS			
State Load Prediction	Stack: Python, SARIMA, NumPy, Matplotlib, Power Query M (Dec '24) - Finalist at Smart India Hackathon 2024; developed ML-based electrical grid forecasting system analyzing 5 years of historical data from 15+ state dispatch centers, achieving 92% prediction accuracy. - Built automated ETL pipeline using Power Query M processing 50+ heterogeneous report formats, reducing data preparation time from 40 hours to 2 hours weekly.		
Autonomous Vehicle Guidance System	Stack: Python, CARLA, Keras, TensorFlow, OpenCV, PPO (Oct '23) - Developed end-to-end autonomous driving system processing 30 FPS camera feeds and LiDAR data through CNN-based perception and PPO reinforcement learning, achieving 85% navigation success rate. - Trained PPO agent over 500+ simulation episodes with custom reward shaping, reducing collision rate by 65% while maintaining traffic rule compliance above 90% in urban scenarios.		
NixOS System Configuration	Stack: Nix, NixOS, Linux, Home Manager, Flakes (Ongoing) - Engineered fully declarative, modular system configuration managing 150+ packages and 20+ services across multiple hosts using NixOS Flakes and Home Manager, enabling atomic rollbacks and reproducible development environments with zero manual state. - Structured across hosts/, modules/, and home/ with hardware abstraction for AMD/NVIDIA/Intel GPU stacks, desktop environments (GNOME, Hyprland), and custom package overlays.		
Personal Website & Portfolio	Stack: HTML, CSS, JavaScript (Ongoing) - Designed and self-hosted a personal portfolio site serving as a central hub for project writeups, short links, and professional presence.		
TECHNICAL SKILLS			
Languages & Tools	Languages: Python, SQL, Bash, Nix, Power Query M, $\LaTeX$ Developer Tools: Git, Docker, Azure, Linux, NixOS, Poetry, CARLA, Qdrant, Arelle, Apache Arrow, DuckDB, FastAPI, Flask		
Frameworks & Competencies	Frameworks: TensorFlow, Keras, PyTorch, LangChain, Scikit-learn, XGBoost, Pandas, NumPy, Polars, Matplotlib, OpenCV Core Competencies: Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning, System Architecture, ETL Pipelines, Data Engineering, MLOps, DevOps, Industrial AI		
SKILLS: Python, SQL, Bash, Nix, Power Query M, $\LaTeX$		CORE: ML, DL, CV, RL, Data Eng, MLOps, System Architecture	